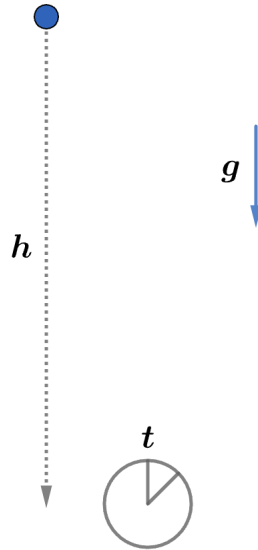


2019B F=ma Exam: Problem 25

Kevin S. Huang



Theory:

$$h = \frac{1}{2}gt^2$$

$$g = \frac{2h}{t^2}$$

Recall when two uncorrelated quantities are multiplied, the relative (fractional) uncertainties add in quadrature. Also, recall the relative uncertainty gets multiplied by $|n|$ when you raise to the n th power. Thus,

$$\frac{\Delta g}{g} = \sqrt{\left(\frac{\Delta h}{h}\right)^2 + \left(\frac{2\Delta t}{t}\right)^2}$$

so the uncertainty in h contributes less than the uncertainty in t . Hence, no uncertainty in t and 10% uncertainty in h minimizes $\Delta g/g$ so the answer is A.